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Work system and work method

Abstract

A work system which performs a work to a structure, and includes an aircraft, and a robot which performs the work to the structure. After the aircraft conveys the robot to the structure while holding the robot, the aircraft releases the robot after the aircraft lowers the robot onto the structure. The robot is released from the aircraft, and then performs the work to the structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This is a U.S. National Phase of International Application No. PCT/JP2021/038649 filed Oct. 19, 2021, which claims the benefit of Japanese Application No. 2020-183354 filed Oct. 30, 2020 and Japanese Application No. 2020-198394 filed Nov. 30, 2020. The disclosure of the prior applications is hereby incorporated by reference herein in its entirety.
TECHNICAL FIELD
(2) The present disclosure relates to a work system and a work method.
BACKGROUND ART
(3) Conventionally, various kinds of work systems are known. As such a work system, there is a delivery system proposed in Patent Document 1, for example.
(4) The delivery system of Patent Document 1 includes a vehicle which stores a package to be delivered to a specific user, and a movable body disposed at a delivery site of the package. The movable body includes a communicating part which transmits and receives given information, and a collection control part which performs a control in which, when the vehicle approaches the delivery site, the movable body moves toward the vehicle from the delivery site and collects the package, and it again moves to the delivery site.
REFERENCE DOCUMENT(S) OF CONVENTIONAL ART
Patent Document
(5) [Patent Document 1] JP2020-083600A
SUMMARY OF THE DISCLOSURE
Problem to be Solved by the Disclosure
(6) In Patent Document 1, it is described that, by having the above-described structure, the storage capacity for the packages can be increased at the vehicle end. However, Patent Document 1 does not take performing a work to a structure into consideration.
(7) Thus, one purpose of the present disclosure is to provide a work system and a work method, capable of performing a work to a structure.
SUMMARY OF THE DISCLOSURE
(8) In order to solve the above-described problem, a work system according to the present disclosure is a work system which performs a work to a structure, and includes an aircraft, and a robot which performs the work to the structure. After the aircraft conveys the robot to the structure while holding the robot, the aircraft releases the robot after the aircraft lowers the robot onto the structure. The robot is released from the aircraft, and then performs the work to the structure.
Effect of the Disclosure
(9) According to the present disclosure, since the robot is conveyed by the aircraft to a structure, is lowered onto the structure from the aircraft and released from the aircraft, and then performs a work to the structure, the work system and the work method capable of performing the work to a structure is provided.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic view illustrating the entire configuration of a work system according to one embodiment of the present disclosure.

(2) FIG. **2** is a schematic side view of a robot provided to the work system according to one embodiment of the present disclosure.

(3) FIG. **3** is a block diagram illustrating a control system of the work system according to one embodiment of the present disclosure.

(4) FIG. **4** is a schematic view illustrating a situation in which the work system according to one embodiment of the present disclosure stores a robot inside an aircraft.

(5) FIG. **5** is a schematic view illustrating a situation in which the work system according to one embodiment of the present disclosure conveys the robots above a structure by the aircraft.

(6) FIG. **6** is a schematic view illustrating a situation in which the work system according to one embodiment of the present disclosure lowers the robot onto the structure from the aircraft.

(7) FIG. **7** is a schematic view illustrating a situation in which, when the work system according to one embodiment of the present disclosure fixes the robot to the structure, the robot extends second robotic arms and grasps a part of a steel tower by second robot hands.

(8) FIG. **8** is a schematic view illustrating a situation in which, when the work system according to one embodiment of the present disclosure fixes the robot to the structure, after it grasps the part of the steel tower by second robot hands of a fixing device, it contracts the second robotic arms.

(9) FIG. **9** is a schematic view illustrating a situation in which the work system according to one embodiment of the present disclosure releases the robot from the aircraft.

(10) FIG. **10** is a schematic view illustrating a situation in which the work system according to one embodiment of the present disclosure performs a work to the structure by using the robots.

(11) FIG. **11** is a schematic view illustrating a situation in which the work system according to one embodiment of the present disclosure recovers the robots by the aircraft.

(12) FIG. **12** is a flowchart illustrating a work method according to one embodiment of the present disclosure.

(13) FIG. **13** is a schematic view illustrating a situation in which the work system according to another embodiment of the present disclosure stores the robot inside the aircraft.

(14) FIG. **14**A is a schematic view illustrating the robot provided to the work system according to one embodiment of the present disclosure, and is a side view of a self-propelled robot which has a similar structure in part to the robot described above.

(15) FIG. **14**B is a schematic view illustrating the robot provided to the work system according to one embodiment of the present disclosure, and is a top view of the self-propelled robot which has a similar structure in part to the robot described above.

(16) FIG. **15** is a schematic view illustrating a robot and an aircraft provided to the work system according to one embodiment of the present disclosure, and is a side view illustrating a situation in which the self-propelled robot having a similar structure in part to the robot described above is attached to another aircraft.

(17) FIG. **16** is an exploded view illustrating one example of a configuration of a mobile robot used for a work system according to another embodiment of the present disclosure.

(18) FIG. **17** is a perspective view illustrating a first configuration and use mode of a self-propelled robot in which the mobile robot of FIG. **16** is configured as a delivery robot.

(19) FIG. **18** is a perspective view illustrating a second configuration and use mode of a self-propelled robot in which the mobile robot of FIG. **16** is configured as a delivery robot.

(20) FIG. **19** is a perspective view illustrating a third configuration and use mode of a self-propelled robot in which the mobile robot of FIG. **16** is configured as a delivery robot.

(21) FIG. **20** is a perspective view illustrating a first configuration and use mode of a high-place walking robot in which the mobile robot of FIG. **16** is configured as a maintenance robot.
(22) FIG. **21** is a perspective view illustrating a second configuration and use mode of a high-place walking robot in which the mobile robot of FIG. **16** is configured as a maintenance robot.
(23) FIG. **22**A is a schematic view illustrating one example of a mode in which the maintenance robot of FIG. **20** moves on the steel tower, and is a side view illustrating an initial state.
(24) FIG. **22**B is a schematic view illustrating one example of a mode in which the maintenance robot of FIG. **20** moves on the steel tower, and is a side view when moving a sticking mechanism of a leg in the moving direction.
(25) FIG. **22**C is a schematic view illustrating one example of a mode in which the maintenance robot of FIG. **20** moves on the steel tower, and is a side view when moving the robot in the moving direction while separating its base from the steel tower.
(26) FIG. **22**D is a schematic view illustrating one example of a mode in which the maintenance robot of FIG. **20** moves on the steel tower, and is a side view when the movement of the maintenance robot in the moving direction is finished.

MODES FOR CARRYING OUT THE DISCLOSURE

(27) Hereinafter, a work system according to one embodiment of the present disclosure is described with reference to the drawings. Note that the present disclosure is not limited by the present disclosure. Further, below, throughout the drawings, the same reference characters are assigned to the same or corresponding elements to omit redundant description.
(28) (Work System **10**)
(29) FIG. **1** is a schematic view illustrating the entire configuration of a work system according to this embodiment. As illustrated in FIG. **1**, a work system **10** includes a VTOL aircraft (Vertical Take-Off and Landing aircraft) **11** as an aircraft, and a robot **30** which performs a work to a steel tower T (structure). Further, the work system **10** is further provided with an interface **90** for an operator P to manipulate the VTOL aircraft **11** and the robot **30** by a remote control.
(30) As illustrated in FIG. **1**, the VTOL aircraft **11** includes a VTOL aircraft body **12**, a propeller **13** attached to an upper part of the VTOL aircraft body **12**, a winch **15** (see FIG. **3**) provided inside the VTOL aircraft body **12**, and a wire rope **16** (see FIG. **4**) which is wound up by the winch **15**. Further, the VTOL aircraft **11** is further provided with a VTOL control device **20** (see FIG. **3**) which controls at least operation of the propeller **13** and the winch **15**.
(31) FIG. **2** is a schematic side view of a robot provided to the work system according to this embodiment. As illustrated in FIG. **2**, the robot **30** includes a robot body **31** and a fixing device **70** which fixes the robot body **31** to the steel tower. The robot **30** further includes a robot control device **60** which controls at least operation of the robot body **50**, the fixing device **70**, and a camera **56** (described later). Note that the robot **30** may be configured to be travelable by itself by providing wheels (for example, four wheels) to a bottom part of a base **32**, and providing a drive of the four wheels inside the base **32**.
(32) The robot body **31** includes the base **32**, a pair of robotic arms **42***a* and **42***b* (first robotic arms) which are attached at their base-end parts to an upper surface of the base **32**, and a pair of robot hands **48***a* and **48***b* (first robot hands) which are provided to tip ends of the pair of robotic arms **42***a* and **42***b* and perform a work to the steel tower T. The pair of robotic arms **42***a* and **42***b* are provided so as to be separated from each other in the left-and-right direction of the robot **30** perpendicular to the front-and-rear direction and the up-and-down direction which are illustrated in FIG. **2**. The robot hand **48***a* is provided to the tip end of the robotic arm **42***a*, and the robot hand **48***b* is provided to the tip end of the robotic arm **42***b*.
(33) Note that, although the structures of the pair of robotic arms **40***a* and **40***b* and the pair of robot hands **48***a* and **48***b* are illustrated in a simplified fashion in FIG. **2** etc., they may have similar structures to a self-propelled robot **120** illustrated in FIGS. **14**A and **14**B. The detailed structure of the self-propelled robot **120** will be described later.

(34) The robot body **31** further includes an auxiliary robotic arm **52** attached at its base-end part to a rear-end part of the upper surface of the base **32**, a display **54** provided to a tip end of the auxiliary robotic arm **52**, and the camera **56** provided to an edge part of the display **54**. The base-end part of the auxiliary robotic arm **52** is provided between the pair of robot hands **48***a* and **48***b* in the left-and-right direction of the robot **30**.
(35) The fixing device **70** includes a pair of robotic arms **72***a* and **72***b* (second robotic arms) attached at their base-end parts to a bottom surface of the base **32**, and a pair of robotic hands **78***a* and **78***b* (second robotic arms) provided to tip ends of the pair of robotic arms **72***a* and **72***b*. The robot hand **78***a* is provided to the tip end of the robotic arm **72***a*, and the robot hand **78***b* is provided to the tip end of the robotic arm **72***b*. The fixing device **70** fixes the robot body **31** to the steel tower T by the pair of robot hands **78***a* and **78***b* grasping (holding) a part of the steel tower T.
(36) Note that, although the structures of the pair of robotic arms **160***a* and **160***b* and the pair of robot hands **68***a* and **68***b* are illustrated in a simplified fashion in FIG. **2** etc., they may have similar structures to the self-propelled robot **120** illustrated in FIGS. **14**A and **14**B. As described above, the detailed structure of the self-propelled robot **120** will be described later.
(37) FIG. **3** is a block diagram illustrating a control system of the work system according to this embodiment. As illustrated in FIG. **3**, the interface **90** includes a VTOL interface **91** for manipulating the VTOL aircraft **11** by a remote control, and a robot interface **96** for manipulating the robot **30** by a remote control.
(38) The VTOL interface **91** includes an interface part **92** for accepting an operational input by the operator P, and a transmission part **93** for transmitting the operational input accepted by the interface part **92** to the VTOL aircraft **11**, as a command value. Further, the robot interface **96** includes an interface part **97** for accepting an operational input by the operator P, and a transmission part **98** for transmitting the operational input accepted by the interface part **97** to the robot **30**, as a command value.
(39) The VTOL control device **20** includes a reception part **22** for receiving the command value transmitted from the transmission part **93** of the VTOL interface **91**. Further, the VTOL control device **20** further includes a propeller control part **24** for controlling operation of the propeller **13** based on the command value received by the reception part **22**, and a winch control part **26** for similarly controlling operation of the winch **15**.
(40) The robot control device **60** includes a reception part **62** for receiving the command value transmitted from the transmission part **98** of the robot interface **96**. Further, the robot control device **60** further includes a camera control part **64** for controlling operation of the camera based on the command value received by the reception part **62**, a robot control part **66** for similarly controlling operation of the robot body **31**, and a fixing device control part **68** for similarly controlling operation of the fixing device **70**.
(41) (Work Mode of Work System **10**)
(42) Next, one example of a mode of a work which the work system **10** performs to the steel tower T is described mainly based on FIGS. **4** to **10**. First, as illustrated in FIG. **4**, the VTOL aircraft **11** and the two robots **30** are prepared, a storage door **17** provided to the VTOL aircraft body **12** is opened, and the robots **30** are stored inside the VTOL aircraft body **12**.
(43) At this time, the robots **30** may be stored inside the VTOL aircraft body **12** in a state where the pair of robotic arms **42***a* and **42***b*, the pair of robotic arms **72***a* and **72***b*, and the auxiliary robotic arm **52** are folded. For example, the robots **30** may be carried into the VTOL aircraft body **12** by manual labor. Alternatively, for example, the robots **30** may be configured to be travelable by themselves (self-propelled) so that they travel by themselves and are stored inside the VTOL aircraft body **12**. At this time, the robots **30** may autonomously travel by themselves, or may travel by themselves by a remote control of the operator P.
(44) FIG. **5** is a schematic view illustrating a situation in which the work system according to this embodiment conveys the robots over a structure by the aircraft. As illustrated in FIG. **5**, the VTOL

aircraft **11** conveys the robots **30** over the steel tower T in a state where it stores the robots **30**. FIG. **5** illustrates a state where the VTOL aircraft **11** hovers over the steel tower T in a state where it stores the robots **30**.

(45) As described above, the VTOL aircraft **11** may autonomously perform that the VTOL aircraft **11** conveys the robots **30** over the steel tower T (in other words, flies and moves from a location where it stores the robots **30** to a location above the steel tower T), and hovers over the steel tower T.

(46) FIG. **6** is a schematic view illustrating a situation in which the work system according to this embodiment lowers the robot onto the structure from the aircraft. Next, as illustrated in FIG. **6**, the VTOL aircraft **11** lowers one of the robots **30** on the steel tower T. In detail, the VTOL aircraft **11** opens a door **12***a* provided to the bottom part of the VTOL aircraft body **12** in the state where it hovers over steel tower T. Then, the wire rope **16** is unwound downwardly from the VTOL aircraft body **12** in a state where one robot **30** is attached to its tip end.

(47) When the VTOL aircraft **11** lowers one robot **30** on the steel tower T as described above, for example, the VTOL aircraft **11** may autonomously perform that the VTOL aircraft **11** hovers over the robot **30**, and the operator P may remotely perform using the interface **90** that the winch **15** unwinds the wire rope **16**.

(48) FIG. **7** is a schematic view illustrating a situation in which, when the work system according to this embodiment fixes the robot to the structure, it extends the second robotic arms and grasps a part of the steel tower by the second robot hands. Further, as illustrated in FIG. **7**, it extends the pair of robotic arms **72***a* and **72***b* of the fixing device **70**, and grasps a part of a main post Ta of the steel tower T (a part of the structure) by the pair of robot hands **78***a* and **78***b* of the fixing device **70**. Note that, for example, additionally or alternatively to the pair of robotic arms **72***a* and **72***b* and the pair of robot hands **78***a* and **78***b*, the fixing device **70** may be provided in a side surface of the base **32** with magnet(s) which can stick to the steel tower T.

(49) FIG. **8** is a schematic view illustrating a situation in which, when the work system according to this embodiment fixes the robot to the structure, it contracts the second robotic arms after it grasps a part of the steel tower by the second robot hands of the fixing device. Then, as illustrated in FIG. **8**, after the pair of robot hands **78***a* and **78***b* of the fixing device **70** grasp the main post Ta, the pair of robotic arms **72***a* and **72***b* are contracted so that the robot body **31** is brought closer to a part of the main post Ta which is grasped by the pair of robotic hands **78***a* and **78***b*. As described above, one robot **30** can be fixed to the steel tower T by the fixing device **70**.

(50) FIG. **9** is a schematic view illustrating a situation in which the work system according to this embodiment releases the robot from the aircraft. As illustrated in FIG. **9**, once it is confirmed that one robot **30** is fixed to the steel tower T by the fixing device **70**, the wire rope **16** is removed from the robot **30** to release the robot **30** from the VTOL aircraft **11**, and then the wire rope **16** is wound up by the winch **15**.

(51) Note that, similarly to the above, the other robot **30** is fixed to the steel tower T by the fixing device **70**, and the other robot **30** is then released from the VTOL aircraft **11**. Note that, after the VTOL aircraft **11** released the robots **30** over the steel tower T, it may close the door **12***a* and leave from above the steel tower T.

(52) The operator P may remotely perform using the interface **90** that the robots **30** are fixed to the steel tower T, and the robots **30** are released from the VTOL aircraft **11**, as described above. Further, the VTOL aircraft may autonomously perform that the VTOL aircraft **11** closes the door **12***a* and leaves from above the steel tower T, as described above.

(53) FIG. **10** is a schematic view illustrating a situation in which the work system according to this embodiment performs a work to the structure by using the robots. As illustrated in FIG. **10**, the robots **30** are each fixed to the steel tower T by the fixing device **70**, and each performs a work to the steel tower T by the pair of robot hands **48***a* and **48***b* while changing the posture of the pair of robotic arms **160***a* and **160***b*.

(54) The work to the steel tower T may be, for example, that a part of the pair of robot hands **48***a* and **48***b* is configured as a screw driver, and a bolt or a nut is fastened to the steel tower T by the screw driver. However, without being limited to this case, other works may be performed to the steel tower T by the pair of robot hands **48***a* and **48***b*.
(55) The operator P may remotely perform the work to the steel tower T by the robots **30** as described above by using the interface **90**. Note that the operator P may remotely operate the robots **30**, while switching the robot **30** to be a target of the remote control among the robots **30**. In such a case, the operator P may remotely operate the robots **30**, for example, while switching an image displayed on a display of the interface **90** to an image captured by the camera **56** of the robot **30** which is the target of remote control among the robots **30**.
(56) Note that a camera for acquiring an image by imaging a situation in which the robots **30** perform the work to the steel tower T may be provided to the VTOL aircraft **11**, and the operator P may perform the remote control using the interface **90** based on the image. Note that, for example, the image may be the entire image of the robots **30** and their circumference, or a local image of a part where one of the robots **30** performs the work to the steel tower T, and its circumference.
(57) Alternatively, a drone for image pick-up may be stored beforehand inside the VTOL aircraft **11**, the VTOL aircraft **11** may release the drone for image pick-up over the steel tower T, and the operator P may perform the remote control using the interface **90** based on the image captured by the drone for image pick-up.
(58) FIG. **11** is a schematic view illustrating a situation in which the work system according to this embodiment recovers the robots by the aircraft. Finally, as illustrated in FIG. **11**, the VTOL aircraft **11** recovers the robots **30** on the steel tower T. Note that, FIG. **11** illustrates a state where the VTOL aircraft **11** recovers one of the robots **30** on the steel tower T, after it recovered the other robot **30**.
(59) In detail, for example, the VTOL aircraft **11** flies over the steel tower T before and after the work to the steel tower T by the robots **30** is finished. Next, while the VTOL aircraft **11** hovers over the steel tower T, the door **12***a* of the VTOL aircraft body **12** is opened. Then, the wire rope **16** is unwound downwardly from the VTOL aircraft body **12**. Next, after the other robot **30** is attached to the tip end of the wire rope **16**, the fixing device **70** of the other robot **30** cancels the fixing to the steel tower T (in detail, the pair of robot hands **78***a* and **78***b* each release the part of the main post Ta), and the wire rope **16** is wound up by the winch **15**, thereby storing the other robot **30** inside the VTOL aircraft **11**.
(60) Note that the attachment of the robot **30** to the tip end of the wire rope **16** may be performed by the pair of robot hands **48***a* and **48***b* of the robot **30** grasping the tip end of the wire rope **16**, and the pair of robot hands **48***a* and **48***b* attaching the tip end of the wire rope **16** to a hook provided to the base **32**.
(61) As described above, the VTOL aircraft **11** recovers the other robot **30** on the steel tower T. Further, similarly to the above, the VTOL aircraft **11** recovers one robot **30** on the steel tower T.
(62) The operator P may remotely perform using the interface **90** that the VTOL aircraft **11** recovers the robots **30** on the steel tower T as described above. Note that, when the tip end of the wire rope **16** is attached to the hook of the base **32** by the pair of robot hands **48***a* and **48***b*, the operator P may also remotely perform this work by using the interface **90**.
(63) (Effects)
(64) Since the work system **10** according to this embodiment conveys the robots **30** over the steel tower T by the VTOL aircraft **11**, lowers the robots **30** on the steel tower T from the VTOL aircraft **11**, releases the robots **30** from the VTOL aircraft **11**, and then performs the work to the steel tower T by the robots **30**, it becomes possible to perform the work to the steel tower T.
(65) Further, in this embodiment, since each robot **30** has the fixing device **70**, the robots **30** can perform the work to this steel tower T stably on the steel tower T.
(66) Further, in this embodiment, the operator P remotely performs using the interface **90** that the VTOL aircraft **11** lowers the robots **30** onto the steel tower T, and the robots **30** perform the work to

the steel tower T. Therefore, as compared with the case of autonomously performing that the VTOL aircraft **11** lowers the robots **30** onto the steel tower T and the robots **30** perform the work to the steel tower T, it becomes possible to securely perform these processes.
(67) (One Example of Work Method)
(68) One example of the work method according to one embodiment of the present disclosure is described based on FIG. **12**. FIG. **12** is a flowchart illustrating the work method according to this embodiment.
(69) As illustrated in FIG. **12**, first, Step S**1** (first step) at which the aircraft and the robot are prepared is performed. Note that, although in the above embodiment the VTOL aircraft **11** is prepared as the aircraft, the two robots **30** are prepared as the robot, and the interface **90** is prepared, Step S**1** may be performed in other forms, without being limited to the case described above.
(70) Next, Step S**2** (second step) at which the robot is attached to or is stored inside the aircraft is performed. Note that, although in the above embodiment the robots **30** are stored inside the VTOL aircraft **11**, Step S**2** may be performed in other forms, without being limited to the case described above.
(71) Further, Step S**3** (third step) at which the robot is conveyed over the structure by the aircraft is performed. Note that, although in the above embodiment the robots **30** are conveyed over the steel tower T by the VTOL aircraft **11**, Step S**3** may be performed in other forms, without being limited to the case described above.
(72) Then, Step S**4** (fourth step) at which the robot is lowered onto the structure from the aircraft is performed. Note that, although in the above embodiment the robots **30** are lowered onto the steel tower T from the VTOL aircraft **11** by using the winch **15** and the wire rope **16**, Step S**4** may be performed in other forms, without being limited to the case described above.
(73) Next, Step S**5-1** at which the robot is fixed to the structure is performed. Note that, although in the above embodiment each robot **30** is fixed to the steel tower T by the pair of robot hands **78***a* and **78***b* of the fixing device **70** grasping the main post Ta of the steel tower T, Step S**5-1** may be performed in other forms, without being limited to the case described above.
(74) Further, Step S**5-2** (fifth step) at which the robot is released from the aircraft is performed. Note that, although in the above embodiment the robots **30** are fixed to the steel tower T by the pair of robot hands **78***a* and **78***b* of the fixing device **70** releasing the main post Ta of the steel tower T, Step S**5-2** may be performed in other forms, without being limited to the case described above.
(75) Then, Step S**6** (sixth step) at which a work to the structure is performed by the robot is performed. Note that, although in the above embodiment the robots **30** perform the work to the steel tower T by the pair of robot hands **48***a* and **48***b*, while changing the posture of the pair of robotic arms **160***a* and **160***b*, Step S**6** may be performed in other forms, without being limited to the case described above.
(76) Finally, Step S**7** at which the robot is recovered by the aircraft is performed. Note that, although in the above embodiment the VTOL aircraft **11** recovers the robots **30** by storing the robots **30** inside the VTOL aircraft **11** using the winch **15** and the wire rope **16**, Step S**7** may be performed in other forms, without being limited to the case described above.
OTHER EMBODIMENTS
(77) A work system according to another embodiment of the present disclosure is described based on FIG. **13**. FIG. **13** is a schematic view illustrating a situation in which the work system according to this embodiment stores the robot inside the aircraft. Note that the work system according to this embodiment is similar to the work system **10** according to the above embodiment, except for the mode in which the robots **30** are stored inside the VTOL aircraft **11**. Therefore, the same reference characters are assigned to the same parts not to repeat similar descriptions.
(78) In this embodiment, as illustrated in FIG. **13**, while the VTOL aircraft **11** hovers over one robot **30**, the door **12***a* provided to the bottom part of the VTOL aircraft body **12** is opened, and the

wire rope **16** wound up by the winch **15** is unwound downwardly from the VTOL aircraft body **12**. Then, as illustrated in this drawing, after one robot **30** is attached to the tip end of the wire rope **16**, this robot **30** is stored inside the VTOL aircraft **11** by the wire rope **16** being wound up by the winch **15**. Further, similarly to the above, the other robot **30** is stored inside the VTOL aircraft **11**.
(79) When the robot **30** is stored inside the VTOL aircraft **11** as described above, for example, the VTOL aircraft **11** may autonomously perform that the VTOL aircraft **11** hovers over the robot **30**, and the operator P may remotely perform using the interface **90** that the winch **15** unwinds and winds up the wire rope **16** and the robots **30** are attached to the tip end of the wire rope **16**.
(80) Note that, as illustrated in FIG. **13**, the VTOL aircraft **11** may convey the robot **30** over the steel tower T, while suspending the robot **30** from the wire rope **16**. Alternatively, the VTOL aircraft **11** may suspend a storage device from the wire rope **16**, store the robot **30** inside the storage device while the storage device is placed on the ground, and release the robot **30** from the storage device over the steel tower T.
(81) (Modifications)
(82) It is apparent for the person skilled in the art that many improvement and other embodiments of the present disclosure are possible from the above description. Therefore, the above description is to be interpreted only as illustration, and it is provided in order to teach the person skilled in the art the best mode to implement the present disclosure. The details of the structures and/or the functions may be changed substantially, without departing from the spirit of the present disclosure.
(83) In the above embodiment, the work system **10** is provided with the VTOL aircraft **11** as the aircraft. However, without being limited to this case, the work system **10** may be provided with, for example, a drone **250** as illustrated in FIG. **15**, a manned aircraft which is operated by a person, or other aircrafts. Note that since other aircrafts have known structures, the detailed description thereof is omitted herein.
(84) In the above embodiment, the VTOL aircraft **11** as the aircraft holds the robot **30** by storing the robot **30** inside the VTOL aircraft **11**. However, without being limited to this case, the drone **250** as illustrated in FIG. **15** may be prepared as the aircraft, and the robot **30** may be held by the aircraft by attaching the robot **30** to a lower part of the drone **250** as illustrated in this drawing.
(85) In the above embodiment, the VTOL aircraft **11** as the aircraft conveys the robot **30** over the steel tower T, and the robot **30** performs the work to the steel tower T. However, without being limited to this case, for example, the aircraft may convey the robot **30** to a structure other than the steel tower T, such as a skyscraper, and the robot **30** may perform the work to this structure. Note that the aircraft may convey the robot **30** to a location near a side part of the structure, instead of above the structure.
(86) (Self-Propelled Robot **120**)
(87) Finally, the detailed structure of the self-propelled robot **120** which has a similar structure in part to the robot **30** (see FIG. **2**) according to the above embodiment is described. FIGS. **14**A and **14**B are schematic views illustrating the robot provided to the work system according to the above embodiment, where FIG. **14**A is a side view of the self-propelled robot which has a similar structure in part to the robot described above, and FIG. **14**B is a top view of the self-propelled robot which has a similar structure in part to the robot described above.
(88) For example, the pair of robotic arms **42***a* and **42***b* and the pair of robotic arms **72***a* and **72***b* which are described in the above embodiment may have similar structures to a pair of robotic arms **160***a* and **160***b* illustrated in FIGS. **14**A and **14**B. Further, the pair of robot hands **48***a* and **48***b* and the pair of robot hands **78***a* and **78***b* which are described in the above embodiment may have similar structures to a pair of robot hands **170***a* and **170***b* illustrated in FIGS. **14**A and **14**B.
(89) As illustrated in FIGS. **14**A and **14**B, a travel cart **130** includes a rectangular parallelepiped cart body **132** and four wheels **134***a*-**134***d* attached to a bottom part of the cart body **132**. The wheels **134***a* and **134***b* are attached to a rear part of the cart body **132** via one of axles, and they are rotated by a control device so that the travel cart **130** travels by itself in the moving direction.

Further, the wheels **134**_c_ and **134**_d_ are attached to a front part of the cart body **132** via the other axle, and they are controlled by the control device so that the travel cart **130** changes the moving direction.
(90) A storing container **222** is provided to a front surface of the cart body **132** via a mounting part **220**. In other words, the storing container **222** is provided in front of the travel cart **130**. The storing container **222** has a rectangular parallelepiped shape which is hollow and elongated in the height direction, where articles G.sub.1′ and G.sub.2′ can be loaded and stored. An opening **224**_a_ is formed entirely in an upper surface of the storing container **222**. A rectangular opening **224**_b_ is formed in a rear surface of the storing container **222**, which extends entirely in the width direction and extends from an upper edge of the storing container **222** to near a center part in the height direction. The openings **224**_a_ and **224**_b_ are formed so as to be connected at a corner of the storing container **222**, where the upper surface and the rear surface of the storing container **222** are connected and which extends in the left-and-right direction.
(91) (Robot Body **150**)
(92) As illustrated in FIGS. **14**A and **14**B, the robot body **150** includes a base part **152** which is provided to the upper part of the travel cart **130** and is swivelable on a rotation axis AX.sub.1′ extending vertically, and the pair of robotic arms **160**_a_ and **160**_b_ coupled at their base-end parts to the base part **152**. Further, the robot body **150** also includes the robot hands **170**_a_ and **170**_b_ which are provided to tip ends of the pair of robotic arms **160**_a_ and **160**_b_, respectively.
(93) (Base Part **152**)
(94) The base part **152** is provided to a rear part of the upper surface of the cart body **132**. The base part **152** has a cylindrical shape, and is provided so that its bottom surface contacts or substantially contacts the rear part of the upper surface of the cart body **132**. A rear end of the base part **152** is located at the same position as the rear surface of the cart body **132** in the front-and-rear direction. Note that the rear end of the base part **152** may be located forward of the rear surface of the cart body **132**. The center line of the base part **152** is located on the center line of the cart body **132** which extends in the front-and-rear direction at the center of the cart body **132** in the left-and-right direction. The base part **152** is swivelable on the rotation axis AX.sub.1′ which extends vertically. In other words, the robot body **150** has a joint part JT.sub.1′ which couples the cart body **132** to the base part **152** swivelably on the rotation axis AX.sub.1′.
(95) (Pair of Robotic Arms **160**_a_ and **160**_b_)
(96) The pair of robotic arms **160**_a_ and **160**_b_ each includes a link **162** and a link **164** which is coupled at its base-end part to a tip-end part of the link **162** via a joint part JT.sub.3′. In a retracted state illustrated in FIGS. **14**A and **14**B, the pair of robotic arms **160**_a_ and **160**_b_ extend along the center line of the self-propelled robot **120** which extends in the front-and-rear direction at the center of the self-propelled robot **120** in the left-and-right direction, and are plane symmetry to each other with respect to a plane parallel to both side surfaces of the cart body **132**. The pair of robotic arms **160**_a_ and **160**_b_ are operable independently, or operable collaboratively with each other.
(97) Base-end parts of the pair of links **162** are coupled to the base part **152** so that they are coaxially pivotable on a rotation axis AX.sub.2′ extending horizontally, and so that they oppose to each other via the base part **152**. In other words, the pair of robotic arms **160**_a_ and **160**_b_ each have a Joint part JT.sub.2′ which couples the base part **152** to the link **162** so as to be rotatable on the rotation axis AX.sub.2′.
(98) The pair of links **162** are extendable and contractible in the longitudinal direction by each having a base-end part **163** and a tip-end-part **163**′ which is extendable and contractible in a direction of projecting from a tip end of the base-end part **163**. Thus, for example, by contracting the pair of robotic arms **160**_a_ and **160**_b_, the pair of robot hands **170**_a_ and **170**_b_ become easier to be inserted into the storing container **122**.
(99) Base-end parts of the pair of links **164** are each coupled to a side part of the tip-end part of the

corresponding link **162**, which is on the side opposing to the other link **162** via a cube-shaped coupling part. The tip-end parts of the pair of links **162** and the base-end parts of the pair of links **164** are each formed in a semicircular shape as seen in the corresponding thickness direction.
(100) The joint part JT.sub.3′ (i.e., the joint part JT.sub.3′ which intervenes between the link **162** and the link **164**) including the coupling part can rotate the link **164** with respect to the link **162** on the rotation axis AX.sub.3′ extending horizontally, and on a rotation axis AX.sub.4′ perpendicular to the rotation axis AX.sub.3′.
(101) In other words, by the coupling part rotating on the rotation axis AX.sub.3′ with respect to the tip-end part of the link **162**, the link **162** is rotatable on the rotation axis AX.sub.3′ integrally with the coupling part. Moreover, by the link **164** rotating on the rotation axis AX.sub.4′ with respect to the coupling part, the link **164** is rotatable on the rotation axis AX.sub.4′.
(102) FIG. **15** is a schematic view illustrating a robot and an aircraft provided to the work system according to this embodiment, and is a side view illustrating a situation in which the self-propelled robot having a similar structure in part to the robot described above is attached to another aircraft.
(103) For example, when the pair of robotic arms **42***a* and **42***b* and the pair of robot hands **48***a* and **48***b* which are described in the above embodiment has similar structures to the pair of robotic arms **160***a* and **160***b* and the pair of robot hands **170***a* and **170***b* which are illustrated in FIGS. **14**A and **14**B, the robot **30** which is described in the above embodiment may be held by the drone **250** by being attached underneath the drone **250** as the aircraft in a mode illustrated in FIG. **15**.
(104) As illustrated in FIG. **15**, the drone **250** includes a drone body **252** and four propellers **154***a*-**154***d* attached to the drone body **252**. For example, the drone **250** is capable of generating electricity by using rotation of the propellers **154***a*-**154***d*.
(105) As illustrated in FIG. **15**, the drone **250** further includes a pair of handles **156***a* and **156***b*. The pair of handles **156***a* and **156***b* are provided to a bottom part of the drone body **252** so as to correspond to the pair of robot hands **170***a* and **170***b* of the self-propelled robot **120** which becomes in the retracted state. After becoming in the retracted state, the self-propelled robot **120** is attached underneath the drone **250** by the robot hand **70***a* grasping the handle **156***a* and the robot hand **70***b* grasping the handle **156***b*.
(106) In FIG. **15**, the self-propelled robot **20** is in a state where the articles G.sub.1′ and G.sub.2′ are stored in the storing container **122**. At this time, the self-propelled robot **120** becoming in the retracted state by piling the pair of links **162** and the pair of links **164** in the height direction in the side view so that the entire center of gravity of the self-propelled robot **120** and the articles G.sub.1′ and G.sub.2′ can be located at a center part of the self-propelled robot **120**.
OTHER EMBODIMENTS
(107) A work system according to another embodiment is described based on FIGS. **16** to **21**, and **22**A and **22**B. Note that a high-place walking robot **30**B illustrated in FIG. **20** is a first modification of the robot **30** described based on FIGS. **1** to **13**, and a high-place walking robot **30**B illustrated in FIG. **20** is a second modification of the robot **30**.
(108) FIG. **16** is an exploded view illustrating one example of a configuration of a mobile robot **1000** used for an unmanned delivery system according to another embodiment of the present disclosure. FIG. **17** is a perspective view illustrating a first configuration and use mode of the self-propelled robot **30**A in which the mobile robot **1000** of FIG. **16** is configured as a delivery robot. FIG. **18** is a perspective view illustrating a second configuration and use mode of the self-propelled robot **30**A in which the mobile robot **1000** of FIG. **16** is configured as a delivery robot. FIG. **19** is a perspective view illustrating a third configuration and use mode of the self-propelled robot **30**A in which the mobile robot **1000** of FIG. **16** is configured as a delivery robot. FIG. **20** is a perspective view illustrating a first configuration and use mode of the high-place walking robot **30**B in which the mobile robot **1000** of FIG. **16** is configured as a maintenance robot. FIG. **21** is a perspective view illustrating a second configuration and use mode of the high-place walking robot **30**B in which the mobile robot **1000** of FIG. **16** is configured as a maintenance robot.

(109) Referring to FIG. **16**, the mobile robot **1000** may be selectively configured as the self-propelled robot **30**A which is a delivery robot specialized in delivery, or the high-place walking robot **30**B which is a maintenance robot specialized in maintenance of a high-rise structure. Below, this is described in detail.
(110) The mobile robot **1000** includes a base unit **310** (corresponding to the base **32** described based on FIGS. **1** to **13**), a robotic-arm part **320** or **330**, and a mobile parts **340** or **350**. In FIG. **10**, the base unit **310** is illustrated at the center, the robotic-arm part **320** (third robotic arm) and a wagon **360** are illustrated at the upper left, the mobile part **340** is illustrated at the lower left, the robotic-arm part **330** is illustrated at the upper right, and the mobile part **350** (second robotic arm) is illustrated at the lower right.
(111) By attaching the robotic-arm part **320** to an upper surface of the base unit **310**, and attaching the mobile part **340** to side surfaces at both ends of the base unit **310**, the self-propelled robot **30**A which is the delivery robot is configured (see FIGS. **17** to **19**). By attaching the robotic-arm part **330** to the upper surface of the base unit **310**, and attaching the mobile part **350** to the side surfaces on both ends of the base unit **310**, the high-place walking robot **30**B which is a maintenance robot is configured (see FIGS. **20** to **21** and FIGS. **22**A to **22**D).
(112) (Base Unit **310**)
(113) The base unit **310** is a part which constitutes the body and chassis of the mobile robot **1000**, and is formed in a shape having a substantially constant thickness and having thin-width parts at both ends in the longitudinal direction. The base unit **310** is provided at the upper surface of the center part of the base unit **310** with a robotic-arm part mounting part **311** where the robotic-arm part **320** or **330** is attached. For example, the robotic-arm part mounting part **311** is formed in a short pillar shape, and is provided to the main body of the base unit **310** so as to be rotatable by a motor (not illustrated) on a rotation axis A**300** perpendicular to the upper surface of the center part of the base unit **310**. The robotic-arm part mounting part **311** is provided so that its upper surface becomes flush with the upper surface of the center part of the base unit **310**.
(114) Further, a mobile part mounting part **312** is provided to each side surface of the thin-width part at each end part of the base unit **310**, and an opening is formed in the mobile part mounting part **312**. End parts **313** of axles to which the mobile part **340** or **350** is coupled are exposed from the openings.
(115) One pair of the two pairs of axles corresponding to the thin-width parts on both ends of the base unit **310** are configured to be steerable, and the one pair of the two pairs of axles are driving axles which are driven by a driving source (not illustrated) and the other pair of axles are driven axles. Note that both of the pairs of axles may be driving axles. The driving source is comprised of a motor, for example.
(116) A battery **328** and a robot controller **1201** are mounted on the base unit **310**. The battery **328** supplies electric power for operating the mobile robot **1000**. The robot controller **1201** is configured similarly to the robot controller **201** of Embodiment 1.
(117) Note that, if crawlers are attached to the base unit **310** as the mobile part **340**C, the base unit **310** is formed to have a thin width throughout its length, and the robotic-arm part mounting part **311** is formed integrally with the main body (inrotatable). Further, one pair of axles are configured to be non-steering axles. Note that, also in this case, the robotic-arm part mounting part **311** may be configured to be rotatable, and the robotic-arm part **320** may be configured to be attachable to the robotic-arm part mounting part **311**.
(118) Further, as for the base unit **310**, if the mobile part **350** (second robotic arm) is attached to the mobile part mounting part **312**, each axle is driven individually by the motor as a base-end link of the robotic arm, while being position-controlled.
(119) (Robotic-Arm Part **320**)
(120) The robotic-arm part **320** is a robotic-arm part which constitutes the self-propelled robot **30**A. In order to handle the package to be delivered, since it is necessary to lift the package to a

certain height, the robotic-arm part **320** is provided with a torso part **321** extending upwardly perpendicular to the upper surface of the robotic-arm part mounting part **311**. A pair of robotic arms **322** are provided to a top part of the torso part **321**. Each robotic arm **322** is comprised of an articulated robotic arm (here, multiple-joint arm). Note that the configuration of the robotic arm is not limited in particular, and it may be a horizontal articulated arm (so-called "SCARA arm") besides the vertical articulated arm. A hand **322***a* is attached to a tip end of the robotic arm **322**. The configuration of the hand **322***a* is not limited in particular. Here, the hand **322***a* is comprised of a suction hand which sucks an object using vacuum. For example, the hand **322***a* may be comprised of a hand which pinches the object from both sides.
(121) A customer's display **323** is provided to the top of the torso part **321**. The customer's display **323** is provided with a customer microphone **324**, a customer speaker **325**, and a field-of-view camera **326**. By these, a conversation between the self-propelled robot **30**A and the addressee (customer) at the address for delivery becomes possible.
(122) Upon delivery, the self-propelled robot **30**A is coupled to the wagon **360** which accommodates the package to be delivered (see FIGS. **17** to **19**). Without traveling by itself, the wagon **360** travels by being pushed or pulled by the self-propelled robot **30**A. Below, a front-and-rear direction in the traveling direction of the wagon **360** is referred to as a front-and-rear direction of the wagon **360**. The wagon **360** is provided with a main body **361** which is comprised of a rectangular parallelepiped box. An interior space of this main body **361** is an accommodation space of the package to be delivered.
(123) The main body **361** has a stepped part **365** which is dented forward in a lower part of a rear end surface. An opening-and-closing door **364** is provided to the rear surface of the main body **361**, above the stepped part **365**. This opening-and-closing door **364** is for taking the package to be delivered in and out of the package accommodation space of the main body **361**.
(124) Wheels **362** are provided to four corners of a bottom part of the main body **361**, respectively.
(125) A pair of coupling parts **361***a* which are comprised of protrusions are provided to both side surfaces of the main body **361**. Coupling holes (not illustrated), each comprised of a hole with a bottom, which accept the pair of hands **332***a* of the self-propelled robot **30**A, respectively, are formed in a rear end surface of the pair of coupling parts **361***a*. The self-propelled robot **30**A is coupled to the wagon **360** by inserting the pair of hands **332***a* into the pair of coupling holes, and sucking the bottoms of the coupling holes. Note that the coupling structure of the coupling part **361***a* to the hand **322***a* is not limited to this structure. The coupling structure may be any structure, as long as it is capable of coupling the coupling parts **361***a* to the hands **322***a*, and, for example, mutually-engagement parts may be provided to the coupling parts **361***a* and the hands **322***a* so that they are coupled together.
(126) The wagon **360** is further provided with a battery **363**. The coupling part **361***a* is provided with a first electric contact (not illustrated) electrically connected to the battery **363**, and the hand **322***a* of the self-propelled robot **30**A is provided with a second electric contact (not illustrated) electrically connected to the battery **328**. When the self-propelled robot **30**A is coupled to the wagon **360**, the first electric contact contacts the second electric contact to allow a flow of electric current so that the battery **328** of the self-propelled robot **30**A is charged by the battery **363** of the wagon **360**. This charging is appropriately performed by a control of the base unit **310** by the robot controller **1201**, as needed. Therefore, a travelable distance of the self-propelled robot **30**A becomes longer than the case where the wagon **360** is not provided with the battery **363**.
(127) (Mobile Part **340**)
(128) The mobile part **340** is comprised of three kinds of traveling parts which propels the mobile robot **1000**.
(129) A first mobile part **340**A is comprised of an indoor tire as a first traveling part. The indoor tire is formed, for example, so that the irregularity of the tread (traveling surface) is comparatively small. The indoor tire is attached to the base unit **310** so that its rotation shaft is coupled to the end

part **313** of the axle of the mobile part mounting part **312** of the base unit **310**.
(130) A second mobile part **340**B is comprised of an outdoor tire as a second traveling part. The outdoor tire is formed so that the irregularity of the tread (traveling surface) is comparatively large. Further, a suspension is attached to the tire. The outdoor tire is attached to the base unit **310** so that its rotation shaft is coupled to the end part **313** of the axle of the mobile part mounting part **312** of the base unit **310**. Further, the suspension is suitably coupled to the base unit **310**.
(131) A third mobile part **340**C is comprised of a crawler (caterpillar) as a third traveling part. The crawler is attached to the base unit **310** so that its drive mechanism is coupled to the end part **313** of the axle of the mobile part mounting part **312** of the base unit **310**.
(132) (Robotic-Arm Part **330**)
(133) The robotic-arm part **330** is a robotic-arm part which constitutes the high-place walking robot **30**B. The robotic-arm part **330** includes a pair of robotic arms **331** (first robotic arms). Each robotic arm **331** is comprised of an articulated robotic arm (here, six-axis robotic arm). A hand **331***a* (first robot hand) is attached to a tip end of the robotic arm **331**. The configuration of the hand **331***a* is not limited in particular. Here, the hand **331***a* is comprised of a suction hand which sucks an object using vacuum. For example, the hand **331***a* may be comprised of a hand which pinches the object.
(134) Since the robotic-arm part **330** requires a long horizontally-extending arm in order to perform maintenance at the heights, the two robotic arms **331** are directly attached to the robotic-arm part mounting part **311** of the base unit **310**. Therefore, the two robotic arms **331** are extendable near and along the upper surface of the base unit **310**. Further, since it is necessary to lift the high-place walking robot **30**B up to the height, the robotic-arm part **330** is configured so that the robotic arms **331** are foldable compactly (see FIGS. **20** and **21**).
(135) The robotic-arm part **330** further includes the field-of-view camera **326**. The field-of-view camera **326** is also directly attached to the robotic-arm part mounting part **311** of the base unit **310**. A microphone and a speaker for cooperation with a field worker and gathering circumference information may also be provided.
(136) (Mobile Part **350**)
(137) The mobile part **350** (second robotic arm) is comprised of two kinds of legs which make the mobile robot **1000** perform a height walk.
(138) A fourth mobile part **350**A is comprised of a short leg as a first leg. The short leg is comprised of a five-axis robotic arm, for example. As for the five-axis robotic arm, a base-end link **354** corresponds to a root part of the leg, and a tip-end part **352** (second robot hand) corresponds to a foot part of the leg. The base-end link **354** is coupled to the end part **313** of the axle of the mobile part mounting part **312** of the base unit **310**. The tip-end part **352** is configured to be twistable to the coupling link. The tip-end part **352** is configured to stick to an object. Here, the tip-end part **352** is provided with an electromagnet so that, by turning on the electromagnet, the tip-end part **352** sticks to a magnetic object, and by turning off the electromagnet, the tip-end part **352** is released from the magnetic object. Therefore, while the tip-end part **352** sticks and is fixed to the object in a state where the twist rotation axis of the tip-end part **352** is parallel to the rotation axis of the base-end link **354**, and the twist rotation of the tip-end part **352** is compliance-controlled, when the base-end link **354** is rotated, the base unit **310** moves in the opposite direction from the rotation direction. Therefore, as will be described later, the high-place walking robot **30**B can walk like a caterpillar (measuring worm).
(139) The fourth mobile part **350**A further includes a hollow fixed cover member **353**. The fixed cover member **353** is fixed to the mobile part mounting part **312** of the base unit **310** so that the base-end link **354** rotatably penetrates therethrough. Therefore, the short leg is attached to the base unit **310**.
(140) A fifth mobile part **350**B is comprised of a long leg as a second leg. The long leg is comprised of a seven-axis robotic arm, for example. Other configurations are similar to those of the fourth mobile part **350**A.

(141) (First Configuration and Use Mode of Self-Propelled Robot **30**A)
(142) Referring to FIG. **17**, in the first configuration of the self-propelled robot **30**A, the robotic-arm part **320** is attached to the robotic-arm part mounting part **311** of the base unit **310**, and the indoor tire of the first mobile part **340**A is attached to each mobile part mounting part **312** of the base unit **310**. Therefore, a delivery robot for indoor traveling is configured as the first configuration of the self-propelled robot **30**A.
(143) The self-propelled robot **30**A is used for conveying packages at a collection-and-delivery base (collection-and-delivery center), for example. In this case, the self-propelled robot **30**A performs the following collection-and-delivery work, for example.
(144) First, the self-propelled robot **30**A coupled itself to the wagon **360** by inserting the pair of hands **322***a* of the pair of robotic arms **322** into the coupling holes of the pair of coupling parts **361***a* of the wagon **360**, and sucking the bottoms of the coupling holes by the hands **322***a*. At this time, the battery **328** of the self-propelled robot **30**A is charged by the battery **363** of the wagon **360**. Further, the front end part of the self-propelled robot **30**A is located in the stepped part **365** of the rear surface of the wagon **360**, and the self-propelled robot **30**A approaches and is coupled to the wagon **360**.
(145) Next, the self-propelled robot **30**A travels by itself to a bulk storage, while pushing and pulling the wagon **360**. Next, the self-propelled robot **30**A stops the suction of the pair of hands **322***a*, pulls the pair of hands **322***a* out of the coupling holes of the pair of coupling parts **361***a* of the wagon **360**, and releases the wagon **360** therefrom. Next, the self-propelled robot **30**A loads the packages into the wagon **360** by itself. That is, the robot which conveys the package is the same as the robot which loads and unloads the package. In detail, the self-propelled robot **30**A uses the pair of robotic arms **322** to open the opening-and-closing door **364** of the wagon **360**, holds the package (not illustrated in FIG. **17**) placed in the bulk storage by the pair of hands **322***a* of the pair of robotic arms **322**, and places it inside the accommodation space of the wagon **360**. Here, the self-propelled robot **30**A performs this process, while rotating the torso part **321**, as needed. When the necessary packages are accommodated in the wagon **360**, the self-propelled robot **30**A closes the opening-and-closing door **364**, couples itself to the wagon **360**, and travels by itself to a given location.
(146) Then, if needed, the self-propelled robot **30**A performs the process in the reversed order to release the wagon **360** therefrom and pick out the package from the wagon **360**.
(147) In the above-described work, if needed, the self-propelled robot **30**A will have a conversation with a person by using the customer's display **323**, the customer microphone **324**, the customer speaker **325**, and the field-of-view camera **326**.
(148) (Second Configuration and Use Mode of Self-Propelled Robot **30**A)
(149) Referring to FIG. **18**, in the second configuration of the self-propelled robot **30**A, the robotic-arm part **320** is attached to the robotic-arm part mounting part **311** of the base unit **310**, and the outdoor tire of the second mobile part **340**B is attached to each mobile part mounting part **312** of the base unit **310**. Therefore, a delivery robot for outdoor traveling is configured as the second configuration of the self-propelled robot **30**A.
(150) Since the self-propelled robot **30**A of the second configuration has the outdoor tires, it is suitably used as the self-propelled robot for delivery which eventually delivers the package to the receiver's address. Other configurations are similar to those of the first configuration of the self-propelled robot **30**A.
(151) (Third Configuration and Use Mode of Self-Propelled Robot **30**A)
(152) Referring to FIG. **19**, in the third configuration of the self-propelled robot **30**A, the robotic-arm part **320** is attached to the robotic-arm part mounting part **311** of the base unit **310**, and the crawler of the third mobile part **340**C is attached to each mobile part mounting part **312** of the base unit **310**. Therefore, a delivery robot for bad terrain traveling is configured as the third configuration of the self-propelled robot **30**A.

(153) Since the self-propelled robot **30**A of the third configuration has the crawlers, it is suitably used as a self-propelled robot for bad terrain delivery which travels a bad terrain and eventually delivers the package to the receiver's address. Other configurations are similar to those of the first configuration of the self-propelled robot **30**A. The bad terrain may be a road during a disaster, and irregular ground, for example. Note that the second configuration of the self-propelled robot **30**A changes the direction by slowing down or stopping one of the crawlers.
(154) (First Configuration and Use Mode of High-Place Walking Robot **30**B)
(155) The high-place walking robot **30**B illustrated in FIG. **20** is the first modification of the robot **30** described based on FIGS. **1** to **13**. Referring to FIG. **20**, in the first configuration of the high-place walking robot **30**B, the robotic-arm part **330** is attached to the robotic-arm part mounting part **311** of the base unit **310** (base). In detail, for example, the pair of robotic arms **331** (first robotic arms) are attached to the robotic-arm part mounting part **311** of the base unit **310** so that they are located symmetrically with respect to the rotation axis A**300**. Then, the field-of-view camera **326** is attached to the robotic-arm part mounting part **311** so that it is located forward of the center of the pair of robotic arms **331**. Further, when a microphone and a speaker for cooperation with a field worker and for gathering circumference information are provided, these are suitably attached to the robotic-arm part mounting part **311** or the field-of-view camera **326**. Moreover, the short leg of the fourth mobile part **350**A is attached to each mobile part mounting part **312** of the base unit **310**. Therefore, the maintenance robot which performs maintenance while walking at the heights is configured as the first configuration of the high-place walking robot **30**B.
(156) The high-place walking robot **30**B of the first configuration is used as follows, for example, similarly to the embodiment described based on FIGS. **1** to **13**.
(157) For example, the high-place walking robot **30**B is conveyed by the drone to a maintenance site, such as a high-rise building (for example, a steel tower). Then, for example, if a magnetic member used as a scaffold (hereinafter, referred to as “the scaffold member”) **371** (for example, a horizontal beam member of the steel tower) exists in the high-rise structure, the high-place walking robot **30**B makes the tip-end part **352** (second robot hand) of each short leg stick to the side surface of the scaffold member **371**. Then, while checking a target object (for example, a wire rod) **372** by the field-of-view camera **326**, it performs necessary maintenance, while sticking to and holding the target object by the pair of hands **331***a* of the pair of robotic arms **331**.
(158) In this case, the high-place walking robot **30**B walks as follows.
(159) For example, in a state where the high-place walking robot **30**B has a little gap with the scaffold member **371**, it sticks the tip-end part **352** of each short leg to the scaffold member **371** while the twist rotation axis of the tip-end part **352** is parallel to the rotation axis of the base-end link **354**, and rotates the base-end link **354** rearward. while the twist rotation of the tip-end part **352** is compliance-controlled. Then, the base unit **310** moves forward and downward by the principle of “parallel link.” When the base unit **310** contacts the scaffold member **371**, the high-place walking robot **30**B moves the tip-end parts **352** of the two pairs of short legs forward, and the sticking and fixing are performed similarly to the above. Then, similarly to the above, when the base-end link **354** is rotated rearward, the base unit **310** moves upward while moving forward, and the moves downward to contact the scaffold member **371**. Subsequently, by repeating this operation, the high-place walking robot **30**B walks like a caterpillar.
(160) Note that, if the scaffold member **371** is not horizontal, the high-place walking robot **30**B can walk like a caterpillar by moving the four short legs forward one by one, while maintaining a so-called “three-point support” state of the short legs.
(161) (Second Configuration and Use Mode of High-Place Walking Robot **30**B)
(162) The high-place walking robot **30**B illustrated in FIG. **21** is the second modification of the robot **30** described based on FIGS. **1** to **13**. Referring to FIG. **21**, in the second configuration of the high-place walking robot **30**B, the robotic-arm part **330** is attached to the robotic-arm part mounting part **311** of the base unit **310**, similarly to the above. Then, the long leg of the fifth

mobile part **350**B is attached to each mobile part mounting part **312** of the base unit **310**. Thus, the maintenance robot which performs maintenance while walking at the heights is configured as the second configuration of the high-place walking robot **30**B.
(163) Since the high-place walking robot **30**B of the second configuration has the longer and thicker long leg than the short leg, it can perform more extensive maintenance.
(164) (Moving Mode of High-Place Walking Robot **30**B on Steel Tower)
(165) Finally, one example of a mode of the high-place walking robot **30**B illustrated in FIG. **20** for moving on the steel tower is described based on FIGS. **22**A-**22**D. Here, one example of a mode in which the high-place walking robot **30**B moves on the scaffold member **371** of the steel tower (corresponding to a cross arm Tb of the steel tower T in the embodiment described based on FIGS. **1** to **13**) in the moving direction illustrated in FIGS. **22**A-**22**D is described.
(166) First, as illustrated in FIG. **22**A, the high-place walking robot **30**B feeds current to an electromagnet **359** provided underneath the base unit **310** to stick the base unit **310** to an upper surface of the scaffold member **371** by an electromagnetic force of the electromagnet **359**. Further, the high-place walking robot **30**B feeds current to the tip-end parts **352** provided to the tip ends of the four mobile parts **350**A to stick the four mobile parts **350**A to the side surfaces of the scaffold member **371** by the electromagnetic forces of the tip-end parts **352**. As described above, the high-place walking robot **30**B can stick itself to the scaffold member **371**. Then, the high-place walking robot **30**B is capable of performing the work to the steel tower in this state by using the pair of robotic arms **331** and the pair of hands **331***a*.
(167) Next, while sticking the base unit **310** to the upper surface of the scaffold member **371**, the high-place walking robot **30**B stops the feeding of current to the four tip-end parts **352** to release the four mobile parts **350**A from the side surfaces of the scaffold member **371**.
(168) Further, as illustrated in FIG. **22**B, while sticking the base unit **310** to the upper surface of the scaffold member **371** by the electromagnetic force of the electromagnet **359**, the high-place walking robot **30**B rotates the four base-end links **354** by the mutually same angle so that the four tip-end parts **352** move in the moving direction by the mutually same distance. Further, as illustrated in this drawing (FIG. **22**B), the high-place walking robot **30**B rotates the joint axes of the four mobile parts **350**A provided to the center part in the longitudinal direction so that the four mobile parts **350**A become longer than in the state illustrated in FIG. **22**A in the side view. Therefore, the four tip-end parts **352** become at the same height as the state illustrated in FIG. **22**A.
(169) Then, as illustrated in this drawing (FIG. **22**B), the high-place walking robot **30**B again feeds current to the four tip-end parts **352** to stick the four mobile parts **350**A to parts of the side surfaces of the scaffold member **371** forward in the moving direction from the state illustrating in FIG. **22**A, by the electromagnetic forces of the tip-end parts **352**.
(170) Next, while sticking the four mobile parts **350**A to the side surfaces of the scaffold member **371** by the electromagnetic forces of the four tip-end parts **352**, the high-place walking robot **30**B stops the feeding of current to the electromagnet **359** to release the base unit **310** from the upper surface of the scaffold member **371**.
(171) Further, as illustrated in FIG. **22**C, while sticking the four mobile parts **350**A to the side surfaces of the scaffold member **371** by the electromagnetic forces of the four tip-end parts **352**, the high-place walking robot **30**B rotates the four base-end links **354** by the mutually same angle so that the four mobile parts **350**A extend vertically (i.e., the four base-end links **354** are located at the same positions as the corresponding tip-end parts **352** in the moving direction). Thus, as illustrated in this drawing (FIG. **22**C), the bottom of the base unit **310** separates from the upper surface of the scaffold member **371**, and the base unit **310** moves in the moving direction forward from the state illustrated in FIGS. **22**A and **22**B.
(172) Then, as illustrated in FIG. **22**D, while sticking the four mobile parts **350**A to the side surfaces of the scaffold member **371** by the electromagnetic forces of the four tip-end parts **352**, the high-place walking robot **30**B rotates the joint axes of the four mobile parts **350**A provided to the

center part in the longitudinal direction so that the four mobile parts **350**A become the same length as the state illustrated in FIG. **22**A in the side view. Therefore, the bottom of the base unit **310** again contacts the upper surface of the scaffold member **371**.
(173) Finally, as illustrated in this drawing (FIG. **22**D), while sticking the four mobile parts **350**A to the side surfaces of the scaffold member **371** by the electromagnetic forces of the four tip-end parts **352**, the high-place walking robot **30**B again feeds current to the electromagnet **359** provided to the bottom of the base unit **310** to again stick the base unit **310** to the upper surface of the scaffold member **371** by the electromagnetic force of the electromagnet **359**. As described above, the high-place walking robot **30**B is capable of moving in the moving direction on the scaffold member **371**.
(174) Note that, in the description above based on FIGS. **22**A-**22**D, the high-place walking robot **30**B moves on the scaffold member **371** which extends horizontally of the steel tower. However, without being limited to this case, the high-place walking robot **30**B is also movable in a similar manner on a main post which extends vertically of the steel tower.
(175) Further, in the description above based on FIGS. **22**A-**22**D, the four mobile parts **350**A stick to the side surfaces of the scaffold member **371** by the electromagnetic force. However, without being limited to this case, a hand which can grasp a part of the steel tower may be provided to the tip end of each of the four legs, and the high-place walking robot **30**B may be fixed to the steel tower by grasping the part of the steel tower by the hands.
(176) (Summary)
(177) In order to solve the above-described problem, the work system according to one embodiment of the present disclosure is a work system which performs a work to a structure, and includes an aircraft, and a robot which performs the work to the structure. After the aircraft conveys the robot to the structure while holding the robot, the aircraft releases the robot after the aircraft lowers the robot onto the structure. The robot is released from the aircraft, and then performs the work to the structure.
(178) According to this configuration, in the work system according to the present disclosure, the robot is conveyed to the structure by the aircraft, the robot is lowered onto the structure from the aircraft, the robot is released from the aircraft, and then the work is performed to the structure by the robot. Therefore, it becomes possible to perform the work to the structure.
(179) The robot may include a robot body and a fixing device which fixes the robot body to the structure. The aircraft may release the robot on the structure after the robot is fixed to the structure by the fixing device.
(180) According to this configuration, since the robot has the fixing device, the robot can perform the work to the structure stably on the structure.
(181) For example, the robot body may include a base, a first robotic arm attached at a base-end part thereof to the base, and a first robot hand which is provided to a tip end of the first robotic arm and performs the work to the structure. The fixing device may be provided to the base.
(182) For example, the fixing device may include a second robotic arm attached at a base-end part thereof to the base, and a second robot hand provided to a tip end of the second robotic arm. The fixing device may fix the robot body to the structure by holding a part of the structure by the second robot hand.
(183) The robot may be movable on the structure by repeating operation in which, after the robot holds a part of the structure by the first robot hand, the robot changes a posture of the first robotic arm and holds another part of the structure by the first robot hand.
(184) According to this configuration, the robot can perform the work to the structure while moving on the structure.
(185) By the base being constituted as a travel cart, the robot may be travelable on the structure by itself.
(186) According to this configuration, the robot can perform the work to the structure while

traveling by itself on the structure.
(187) The work system may further include an interface for an operator to remotely operate the aircraft and the robot. The operator may remotely perform, using the interface, at least one of that the aircraft lowers the robot onto the structure and that the robot performs the work to the structure.
(188) According to this configuration, as compared with the case of autonomously performing that the aircraft lowers the robot onto the structure and the robot performs the work to the structure, it becomes possible to securely perform these processes.
(189) For example, the robot may include a plurality of robots, and the operator may remotely perform, using the interface, that the plurality of robots perform the work to the structure, while switching the robot to be remotely operated among the plurality of robots.
(190) For example, the aircraft may include a storage device which stores the robot, and convey the robot to the structure while storing the robot inside the storage device.
(191) For example, the aircraft may be a drone.
(192) The robot may include three assembly units comprised of the first robotic arm, the base, and a mobile part which moves the robot, and the first robotic arm may be attached to an upper surface of the base, and the mobile part may be attached to a side surface of the base.
(193) According to this configuration, the robot can be assembled easily.
(194) The base may be configured so that a third robotic arm including a torso part extending perpendicularly from an upper surface of the base, and the first robotic arm which is directly attached to the upper surface of the base and is extendable near and along the upper surface of the base, are selectively attached to the upper surface, and a traveling part which propels the robot, and the second robotic arm which makes the robot walk at a high location, are selectively attached to the side surface.
(195) According to this configuration, by attaching the third robotic arm to the upper surface of the base and attaching the traveling part to the side surface of the base, for example, a delivery robot can be configured. Alternatively, by attaching the fourth robotic arm part to the upper surface of the base and attaching the second robotic arm to the side surface of the base, for example, a high-place walking robot for maintenance of a high-rise structure can be configured.
(196) In order to solve the above-described problem, the work method according to one embodiment of the present disclosure is a work method which performs a work to a structure, and includes a first step in which an aircraft and a robot that performs the work to the structure are prepared, a second step in which, after performing the first step, the robot is held by the aircraft, a third step in which, after performing the first and second steps, the robot is conveyed to the structure by the aircraft, a fourth step in which, after performing the first to third steps, the robot is lowered onto the structure from the aircraft, a fifth step in which, after performing the first to fourth steps, the robot is released from the aircraft, and a sixth step in which, after performing the first to fifth steps, the work is performed to the structure by the robot.
(197) According to this configuration, in the work method according to the present disclosure, the robot is conveyed to the structure by the aircraft, the robot is lowered onto the structure from the aircraft, the robot is released from the aircraft, and then the work is performed to the structure by the robot. Therefore, it becomes possible to perform the work to the structure.

Claims

1. A work system that performs a work on a structure, comprising: an aircraft; and a robot that is configured to perform the work on the structure, wherein, after the aircraft conveys the robot to the structure while holding the robot, the aircraft is configured to release the robot after the aircraft lowers the robot onto the structure, the robot includes a robot body and a fixing device that is configured to fix the robot body to the structure, the aircraft is configured to release the robot on the structure after the robot is fixed to the structure by the fixing device, the robot body includes a

base, a first robotic arm attached at a base-end part thereof to the base, and a first robot hand that is provided at a tip end of the first robotic arm and is configured to perform the work on the structure, the fixing device is provided to the base, the fixing device includes a second robotic arm attached at a base-end part thereof to the base, and a second robot hand provided at a tip end of the second robotic arm, and the fixing device is configured to fix the robot body to the structure by grasping and clenching a part of the structure with the second robot hand.

2. The work system of claim 1, wherein the robot is configured to move on the structure by repeating an operation in which, after the robot holds a part of the structure by the first robot hand, the robot changes a posture of the first robotic arm and holds another part of the structure by the first robot hand.

3. The work system of claim 1, wherein, the base is configured as a travel cart, and the robot is configured to travel on the structure by itself.

4. The work system of claim 1, further comprising an interface for an operator to remotely operate the aircraft and the robot, wherein the operator remotely performs, using the interface, at least one operation of controlling the aircraft to lower the robot onto the structure and controlling the robot to perform the work to on the structure.

5. The work system of claim 4, wherein the robot includes a plurality of robots, and wherein the operator remotely controls, using the interface, the plurality of robots to perform the work on the structure, and the interface is configured to switch the robot to be remotely operated from among the plurality of robots.

6. The work system of claim 1, wherein the aircraft includes a storage device that stores the robot, and the aircraft is configured to convey the robot to the structure while storing the robot inside the storage device.

7. The work system of claim 1, wherein the aircraft is a drone.

8. The work system of claim 1, wherein the robot includes three assembly units comprised of the first robotic arm, the base, and a mobile part that moves the robot, and wherein the first robotic arm is attached to an upper surface of the base, and the mobile part is attached to a side surface of the base.

9. The work system of claim 8, wherein the base is configured so that: a third robotic arm including a torso part extending perpendicularly from an upper surface of the base, and the first robotic arm that is directly attached to the upper surface of the base and is extendable near and along the upper surface of the base, are selectively attached to the upper surface, and a traveling part that propels the robot, and the second robotic arm that makes the robot walk at a high location, are attached to the side surface.

10. A work method that performs a work on a structure with an aircraft and a robot, wherein the robot includes a robot body and a fixing device that is configured to fix the robot body to the structure, the aircraft is configured to release the robot on the structure after the robot is fixed to the structure by the fixing device, the robot body includes a base, a first robotic arm attached at a base-end part thereof to the base, and a first robot hand that is provided at a tip end of the first robotic arm and is configured to perform the work on the structure, the fixing device is provided to the base, the fixing device includes a second robotic arm attached at a base-end part thereof to the base, and a second robot hand provided at a tip end of the second robotic arm, and the fixing device is configured to fix the robot body to the structure by grasping and clenching a part of the structure with the second robot hand, the work method comprising the steps of: a first step in which an aircraft and a robot that performs the work to-on the structure are prepared; a second step in which, after performing the first step, the robot is held by the aircraft; a third step in which, after performing the first and second steps, the robot is conveyed to the structure by the aircraft; a fourth step in which, after performing the first to third steps, the robot is lowered onto the structure from the aircraft; a fifth step in which, after performing the first to fourth steps, the robot is released

from the aircraft; and a sixth step in which, after performing the first to fifth steps, the work is performed on the structure by the robot.